

VENKAT DASARI

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EDUCATION

University of Massachusetts Amherst

Master of Science in Data Science

Amherst, MA

May 2026

Coursework: Data Engineering, Natural Language Processing, Research Design, Data-driven Storytelling

Thesis: Examining Epistemic Evidentiality in Political Discourse using LLMs

Gandhi Institute of Technology and Management

Bachelor of Technology in Computer Science and Engineering

Visakhapatnam, India

June 2023

TECHNICAL SKILLS

Languages: Python, R, SQL, JavaScript, HTML, CSS

AI/ML: NumPy, Pandas, Scikit-learn, TensorFlow, HuggingFace, vLLM, Ollama, LangChain, RAG, Prompt Engineering

Visualization: Streamlit, Shiny, React, D3, Tableau

Databases: MySQL, MongoDB, Snowflake, ChromaDB

Other: FastAPI, Docker, dbt, Git, Google Cloud Platform (GCP)

WORK EXPERIENCE

Research Assistant

University of Massachusetts Amherst

November 2024 – Present

Amherst, MA

- Led data collection and built **NLP pipelines in Python using HuggingFace and vLLM** for sentiment, ideology, and citation stance classification across 500+ political books; automated manual labeling and **cut annotation effort by 70%**
- Designed statistical validation protocols to ensure accuracy in ideological constructs and category structures; engineered task-specific prompts to ensure consistency, achieving 85% cross-model label agreement
- Designed and benchmarked a **RAG system over 400k+ text segments**; Optimized retriever-generator configuration, **achieving Recall@20 of 0.85 and reduced hallucination rate by 18%**
- Built and maintaining **interactive dashboards using Streamlit, Shiny, React and D3** covering 500+ books and 300+ authors by ideology, timeline, topics, and citation networks

NLP Consultant

Institute for Social Science Research, UMass

October 2025 – May 2026

Amherst, MA

- Advised **15+ research projects** through presentations and one-on-one consultations, supporting initial scoping, study design, implementation, model evaluation and delivery
- Created reusable tutorials, code templates, and documentation for text-as-data workflows, **reducing new-project onboarding time by half** and standardizing best practices institute-wide
- Led **hands-on R and Python workshops** for **40+ participants** on quantitative text analysis, including pre-processing, statistical/topic modeling, and LLM-based methods

Operations & Technology Analyst

The Bharat Scouts & Guides

January 2024 – June 2024

Rajahmundry, India

- Produced **campaign performance reports using Tableau and Excel** that informed outreach strategy and contributed to a **125% increase in membership** over 3 months
- Built and maintained a database of **800+ member profiles**, improving data quality and standardizing recordkeeping across the organization
- Coordinated cultural, service, and adventure programs in collaboration with Indian Railways and local public offices

SELECTED PROJECTS

IdeologyRAG ([Link](#))

January 2026 – May 2026

- Designed a comparative RAG system over 400k+ text segments, benchmarking 4 embedding models × 4 chunking strategies × 3 retrieval methods and 4 open-source LLMs against a self-constructed, 912-question verified QA dataset
- Achieved Recall@20 of 0.85 using Qwen3-Embedding-4B with 1,000-character paragraph-level chunking; deployed Qwen2.5-14B as generator via independent LLM-judge evaluation with 81% correctness and 94% faithfulness

Examining Epistemic Evidentiality in Political Discourse using LLMs

September 2025 – March 2026

- Built a human-in-the-loop LLM annotation pipeline using Gemma-3-27B, Llama-3.3-70B, and Phi-3.5-Mini for evidentiality classification in 50 U.S. political books and analyzed how political actors use information sources to construct credibility
- Evaluated zero-shot, few-shot, and fine-tuning (LoRA/PEFT) configurations, improving macro recall from 0.43 to 0.57 and macro F1 from 0.40 to 0.52 on a limited human-labeled gold standard

US-China Tech Policy Discourse Analysis

January 2026 – Present

- Orchestrated an end-to-end regulations.gov scraping pipeline collecting 26,000+ public comments across 300+ dockets, handling comment+attachment extraction, text parsing, pagination, and rate-limiting; Executed an LLM annotation pipeline using Gemma-3-27B to classify stance, argument type, and tone on messy, unstructured government filings
- Apply multivariate regression analysis to test hypotheses on U.S. firm lobbying behavior, modeling how competitive position vis-à-vis China, control of firm-specific assets, and supply chain embeddedness jointly shape corporate political strategy on export controls

PUBLICATIONS

- Ka Zeng and **Venkat Dasari**. "Navigating the U.S.-China Tech War: Corporate Lobbying on U.S. Export Controls." Accepted for presentation at the American Political Science Association (APSA) 122nd Annual Meeting & Exhibition, 2026.
- S. S. P. Naresh, R. Talasu, **D. S. S. P. Venkat**, S. K. Korada and B. K. Mohanta, "Real Time Vehicle Tracking using YOLO Algorithm," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-5, doi: 10.1109/ICCCNT56998.2023.10307265.

ACHIEVEMENTS & EXTRACURRICULARS

- **Outstanding Student Award**, *Data Analytics Department at UMass Amherst*, 2026
- **Volunteer Team Lead**, *MIT's GenAI Global*, 2026
- **Volunteer Data Consultant**, *Five College DataFest 2025 (American Statistical Association)*, Advised student teams on statistics, R/Python troubleshooting, and presentation strategy, 2025
- **All India Rank (AIR) of 20**, *National Engineering Olympiad 6.0*, 2023
- National-level esports finalist representing Hex Esports and Minus 40 Club in PUBG Mobile tournaments, 2022
- Active Rover at The Bharat Scouts and Guides, developing teamwork, leadership, service and adventure skills, 2015 – 2024